STUDENT SUCCESS / PREDICTIVE SUPPORT

STUDENT EXAM RESULTS CAN BE PREDICTED BEFORE THE FINAL TEST

A DECISION-SUPPORT PROJECT FOR EARLIER, FAIRER INTERVENTION

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THE PROJECT / PURPOSE

The project turns student signals into an early-warning system

Act before the final exam—while there’s time to help.

01 Predict earlier

Estimate final exam performance before the exam happens.

02 Find support needs

Help teachers identify students who may need support.

03 Clarify the path

Give students a clearer path to improve.



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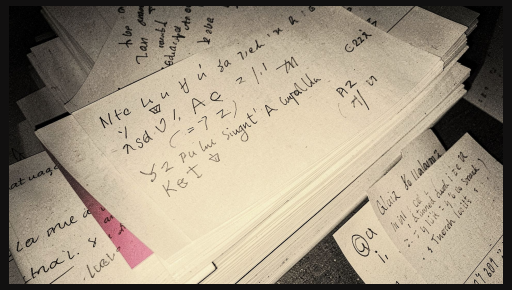
INPUTS / STUDENT SIGNALS

The prediction starts with the right student signals



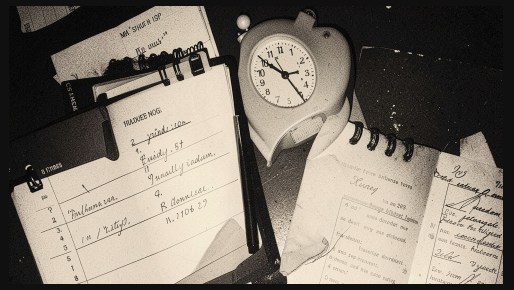
Participation

Attendance and participation.



Performance

Assignment and quiz results.



Preparation

Study time and previous results.



Context

Access to learning resources and support.

USE SIGNALS THAT DESCRIBE LEARNING CONTEXT, NOT LABELS THAT DEFINE A STUDENT.

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PIPELINE / FROM RECORDS TO FEATURES

The data pipeline turns raw records into model-ready features

Each step makes the evidence more consistent, inspectable and usable.

01

Collect

Inspect the dataset and its sources.

02

Clean

Resolve missing, inconsistent or duplicate values.

03

Prepare

Encode categories and scale numeric fields where needed.

04

Split

Separate training data from test data.

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MODELS / BASELINE TO BENCHMARK

Three models create a useful baseline and a stronger benchmark

Compare interpretability and flexibility before making a choice.

01 / BASELINE

Logistic regression

Provides an interpretable starting point for classifying likely outcomes.

02 / RULES

Decision tree

Turns the prediction into easy-to-explain branching rules.

03 / BENCHMARK

Random forest

Tests whether combining trees captures stronger non-linear patterns.

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OUTPUT / DECISION SUPPORT

The output should support action, not just a score

A prediction matters only when it helps a teacher and student decide what to do next.

Outcome

Classify likely outcomes such as pass or at-risk.

Confidence

Show how strongly the model supports each prediction.

Signals

Surface the factors most associated with the result.

Next action

Keep teacher judgment in the loop.

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Evaluation asks whether the model is useful and fair

Different measures reveal different kinds of error. No single score is enough.

Measure	What it tells us	Question to ask
Accuracy	Overall correctness	How often is the classification right?
Precision	Reliability of an at-risk flag	How many flagged students truly need support?
Recall	Coverage of students who need support	How many at-risk students did we find?
F1-score	Balance between precision and recall	Is the trade-off acceptable?
Confusion matrix	Where predictions fail	Which errors carry the greatest cost?

EVALUATION TARGET: USEFULNESS, ERROR COST, AND FAIRNESS ACROSS STUDENT GROUPS

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RISK / RESPONSIBLE USE

The biggest risk is a confident prediction built on weak data

A polished model can still be wrong when the evidence is incomplete, biased or interpreted without context.

Sample

Small or biased samples can mislead the model.

Context

Missing data can hide a student’s real circumstances.

History

Historical outcomes may encode existing inequality.

Safeguard

Use predictions to guide support, never to label a student permanently.

PRIVACY, CONSENT AND HUMAN REVIEW ARE PART OF THE SYSTEM DESIGN.

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INTERFACE / TEACHER DASHBOARD

A teacher dashboard makes the result usable

The interface should organize attention, explain the signal and make the next conversation easier.

Urgency
Group at-risk students by urgency.

Evidence
Show key contributing signals beside each prediction.

Follow-up
Record recommended actions and notes.

Filters
Sort simply by class, subject and time period.



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NEXT ITERATION / EVIDENCE AND FEEDBACK

The project can improve through better evidence and feedback

Improvement is a loop: strengthen the data, test the model, check its fairness and learn from the support that follows.

01

Represent

Collect more representative data.

02

Compare

Compare models with cross-validation.

03

Check

Test fairness across student groups.

04

Learn

Measure whether interventions improve outcomes.

THE NEXT ACTION IS BETTER EVIDENCE, NOT MORE CONFIDENCE.

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